

WTW 158 Worksheet 8 on Using Derivatives

A. TEXTBOOK PROBLEMS

p283 no 27, 33, 39, 49, 55, 61

p 291 no 5, 13, 21

p 300 no 5, 9, 13a and b, 19, 29

B. THEORY QUESTIONS

1. Complete the following definitions

(a) A critical number of a function f is a number c

(b) A function f is increasing on an interval I if

2. State the following theorems:

(a) The Extreme Value Theorem

(b) Fermat's Theorem

(c) The Mean Value Theorem

(d) The Increasing/Decreasing Test

(e) The Concavity Test

3. Write out the proof of the following theorem (The Decreasing/Increasing Test):

If $f' > 0$ on an interval I then f is increasing on that interval.

C. TEST QUESTIONS

1. Do the following functions have critical numbers in $x = 0$?

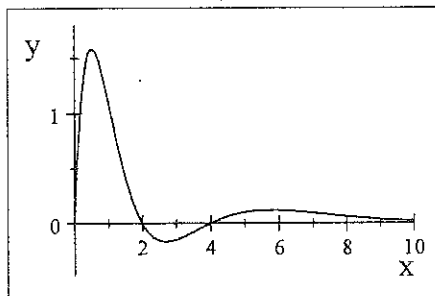
(i) $f(x) = \sqrt{x}$ (ii) $f(x) = \ln x$

2. Say for each of the following statements whether it is true or not. If not an example to illustrate why it is false.

(i) If f has a local extreme in $x = c$ then $f'(x) = 0$

(ii) If $f'(x) = 0$ then f has a local extreme in $x = c$.

3. Consider the derivative graph of the derivative function f'



(i) Where is f increasing? (ii) Where does f have a local maximum? (iii) Where is f concave up?

4. Consider $f(x) = \frac{\ln x}{x}$

(i) Does the Closed Interval Method apply to the function f on $[1, 6]$?

(ii) Find the absolute extremes on $[1, 6]$.

D. SELECTED ANSWERS

A. See textbook.

B. 1., 2. and 3. See textbook.

C. 1. (a) Yes, $x = 0$ is in the domain of the function and the derivative does not exist at $x = 0$. (b)

No, $x = 0$ is not in the domain of f .

2. (i) False, f has to be differentiable, eg $f(x) = |x|$ in $x = 0$.

(ii) False, eg $f(x) = x^3$ in $x = 0$.

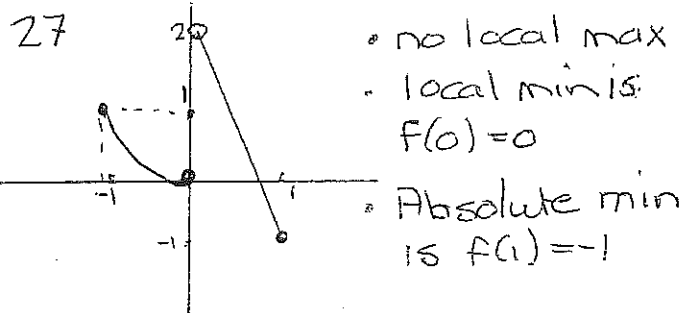
3. (i) $(0, 2) \cup (4, 10)$ (ii) $x = 2$ (iii) $(0, 1) \cup (3, 6)$

4. (i) Yes, f is continuous on $[1, 6]$ (ii) $f(1) = 0$ is the absolute minimum, $f(e) = \frac{1}{e}$ (0, 37) is the local maximum.

absolute

A Textbook Problems

p 283: 27, 33, 39, 49, 55
and 61 (Unit 4.1)



33. $g(t) = t^4 + t^3 + t^2 + 1$
 $\Rightarrow g'(t) = 4t^3 + 3t^2 + 2t$
 $= t(4t^2 + 3t + 2) = 0$
 $\Rightarrow t = 0$ as $4t^2 + 3t + 2 > 0$
 The only critical number is 0.

39. $F(x) = x^{4/5}(x-4)^2$
 $F'(x) = x^{4/5} \cdot 2(x-4) + (x-4)^2 \cdot \frac{4}{5}x^{-1/5}$
 $= \frac{1}{5}x^{-1/5}(x-4)[5x \cdot 2 + (x-4) \cdot 4]$
 $= \frac{(5x-4)(14x-16)}{5x^{1/5}}$
 $= \frac{2(x-4)(7x-8)}{5x^{1/5}}$

Critical numbers:

$F'(x) = 0 \Rightarrow x = 4, x = \frac{8}{7}$
 $F'(0)$ does not exist $\therefore x = 0$

49. $f(x) = 2x^3 - 3x^2 - 12x + 1$ $[-2, 3]$
 $f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2)$
 $= 6(x+1)(x-2) = 0$
 $\Leftrightarrow x = -1, x = 2$
 $\therefore f(-2) = -3, f(-1) = 8$
 $f(2) = -19, f(3) = -8$
 $f(-1) = 8$ Absolute max
 $f(2) = -19$ Absolute min

55. $f(t) = t - \sqrt[3]{t}$ $[-1, 4]$
 $f'(t) = 1 - \frac{1}{3}t^{-2/3} = 1 - \frac{1}{3t^{2/3}}$
 $f'(t) = 0 \Leftrightarrow 1 = \frac{1}{3t^{2/3}} \Rightarrow t^{2/3} = \frac{1}{3}$
 $t = \pm \left(\frac{1}{3}\right)^{3/2} = \pm \frac{1}{\sqrt{27}} = \pm \frac{1}{3\sqrt{3}} = \pm \frac{\sqrt{3}}{9}$
 $f'(t)$ does not exist if $t = 0$
 $f(-1) = 0, f(0) = 0$
 $f\left(\frac{1}{3\sqrt{3}}\right) = \frac{2\sqrt{3}}{9} \approx 0.3849$
 $f\left(\frac{1}{3\sqrt{3}}\right) = -\frac{2\sqrt{3}}{9}$
 $f(4) = 4 - \sqrt[3]{4} \approx 2.413$
 So $f(4) = 4 - \sqrt[3]{4}$ is Absolute max
 $f\left(\frac{\sqrt{3}}{9}\right) = -\frac{2\sqrt{3}}{9}$ Absolute min

61. $f(x) = \ln(x^2 + x + 1)$, $[-1, 1]$
 $f'(x) = \frac{2x+1}{x^2+x+1} = 0 \Rightarrow x = -\frac{1}{2}$
 Since $x^2 + x + 1 > 0$ for $x \in \mathbb{R}$
 $D_f = D_{f'}$
 $f(-1) = \ln 1 = 0$
 $f(1) = \ln 3 \approx 1.10$
 $f\left(-\frac{1}{2}\right) = \ln \frac{3}{4} \approx -0.29$
 Absolute max is $f(1) = \ln 3$
 Absolute min is $f\left(-\frac{1}{2}\right) = \ln \frac{3}{4}$

p 291 no 5, 13, 21
Unit 4.2

5. $f(x) = 2x^2 + 4x + 5$, $[-1, 3]$
 f is a polynomial. So f is continuous and diffb. on $\mathbb{R}$
 Hence: f is continuous on $[-1, 3]$ and diffb. on $(-1, 3)$
 Since $f(-1) = f(3) = 11$
 we can use Rolle's theorem

(2)
 $f'(c) = 4c - 4$ and $f'(c) = 0$
 $\Rightarrow c = 1 \in (-1, 3)$ and Rolle's theorem is satisfied.

13. $f(x) = e^{-2x}$, $[0, 3]$
 • f is continuous on $[0, 3]$
 • f is diffb. on $(0, 3)$
 so f satisfies the conditions of the MVT.

$$(f'(x) = e^{-2x} \cdot -2)$$

$$\text{so } f'(c) = \frac{f(3) - f(0)}{3 - 0}$$

$$\Rightarrow -2e^{-2c} = \frac{e^{-6} - 1}{3}$$

$$\Rightarrow e^{-2c} = \frac{1}{6}(e^{-6} - 1)$$

$$\Rightarrow -2c = \ln\left[\frac{1}{6}(e^{-6} - 1)\right]$$

$$\Rightarrow c = -\frac{1}{2} \ln\left[\frac{1}{6}(e^{-6} - 1)\right]$$

21. Let $f(x) = x^3 - 15x + c$
 for x in $[-2, 2]$.

(We assume that f has two roots and then prove that this assumption is impossible.)

If a and b are real roots in $[-2, 2]$ with $a < b$, then $f(a) = f(b) = 0$. Verify that you can use Rolle which implies that there exists

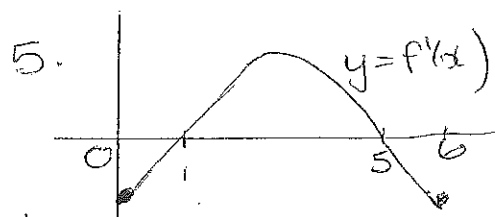
a $c \in (a, b)$ such that $f'(c) = 0$. Now $f'(c) = 3c^2 - 15$.

We know that $|c| < 2$
 so $c^2 < 4$

Then $f'(c) = 3c^2 - 15$
 $< 3(4) - 15 = -3 < 0$

This contradicts $f'(c) = 0$
 so f has at most one real root

p 300: 5, 9, 13a, 13b, 19, 29
 Unit 4.3



(a)
 • $f' > 0$ on $(1, 5)$ so f is increasing on $(1, 5)$
 • $f' < 0$ on $(0, 1)$ and $(5, 6)$ so f is decreasing on $(0, 1)$ and $(5, 6)$

(b) f has a local min $f(1)$ at $x=1$
 f has a local max $f(5)$ at $x=5$

9. $f(x) = x^3 - 3x^2 - 9x + 4$

$$f'(x) = 3x^2 - 6x - 9$$

$$= 3(x+1)(x-3)$$

a) f inc. dec. inc.
 -1 3 since
 f' + - +
 -1 3

b) Local max is $f(-1) = 9$
 Local min is $f(3) = -23$

c) $f''(x) = 6x - 6 = 6(x-1)$

$f''(x) = 0 \Rightarrow x = 1$

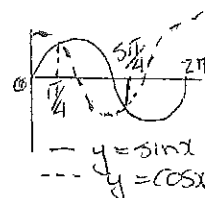
f'' - +
 1
 f is concave up on $(1, \infty)$
 f is concave down on $(-\infty, 1)$
 Point of inflection at $(1, -7)$

13. $f(x) = \sin x + \cos x$, $x \in [0, 2\pi]$

$$f'(x) = \cos x - \sin x$$

$$f''(x) = -\sin x - \cos x$$

a) $f'(x) = 0$
 $\Rightarrow \cos x = \sin x$
 $\Rightarrow x = \pi/4, 5\pi/4$



③

$$f'(x) > 0 \Rightarrow \cos x > \sin x$$

$$f'(x) < 0 \Rightarrow \cos x < \sin x$$

$$f' \begin{array}{c} >0 & <0 & >0 \\ 0 & \pi/4 & 5\pi/4 & 2\pi \end{array}$$

f incr. on $(0, \pi/4)$ and on $(5\pi/4, 2\pi)$

f decr. on $(\pi/4, 5\pi/4)$

b) Local max is $f(\pi/4) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

Local min is $f(5\pi/4) = -\sqrt{2}$

~~$f''(x) = -\sin x - \cos x = 0$~~
(Leave) $\Rightarrow \sin x = -\cos x$

$f''(x) > 0 \Rightarrow \sin x < -\cos x$

$f''(x) < 0 \Rightarrow \sin x > -\cos x$

$\Rightarrow \sin x > -\cos x$

$f'' \begin{array}{c} <0 & >0 & <0 \\ 0 & 3\pi/4 & 7\pi/4 & 2\pi \end{array}$

f is concave downward on $(0, 3\pi/4)$ and on $(7\pi/4, 2\pi)$

f is concave upward on $(3\pi/4, 7\pi/4)$

Inflection points $(3\pi/4, 0), (7\pi/4, 0)$

15. $f(x) = e^{2x} + e^{-x}$

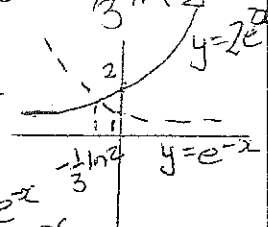
$f'(x) = 2e^{2x} - e^{-x}$

$f''(x) = 4e^{2x} + e^{-x}$

a) $f'(x) = 0 \Rightarrow 2e^{2x} - e^{-x} = 0$
 $\Rightarrow 2e^{2x} = e^{-x} \Rightarrow \ln(2e^{2x}) = \ln e^{-x}$
 $\Rightarrow \ln 2 + \ln e^{2x} = \ln e^{-x}$
 $\Rightarrow \ln 2 + 2x = -x$

$\Rightarrow 3x = -\ln 2 \Rightarrow x = -\frac{1}{3} \ln 2$

$f' \begin{array}{c} <0 & >0 \\ -\frac{1}{3} \ln 2 \end{array}$



$f'(x) > 0 \Rightarrow 2e^{2x} > e^{-x}$

$f'(x) < 0 \Rightarrow 2e^{2x} < e^{-x}$

f is increasing on $(-\frac{1}{3} \ln 2, \infty)$
 f is decreasing on $(-\infty, -\frac{1}{3} \ln 2)$

(b) $f \begin{array}{c} \text{decr.} & \text{incr.} \\ -\frac{1}{3} \ln 2 \end{array}$

Local minimum $f(-\frac{1}{3} \ln 2) = 2^{-2/3} + 2^{1/3}$

(c) $f''(x) = 0 \Rightarrow 4e^{2x} + e^{-x} = 0$

$\Rightarrow 4e^{2x} = -e^{-x}$

$\Rightarrow x \in \emptyset$ No solution

Since $4e^{2x} > 0$ and $-e^{-x} < 0$

But $f''(x) = 4e^{2x} + e^{-x} > 0$ for $x \in \mathbb{R}$. So f is concave upward on $(-\infty, \infty)$

19. $f(x) = x^5 - 5x + 3$

$f'(x) = 5x^4 - 5 = 5(x^2 - 1)(x^2 + 1)$
 $= 5(x-1)(x+1)(x^2+1)$

First Derivative Test

$f'(x) < 0 \Rightarrow 5(x-1)(x+1)(x^2+1) < 0$

$\Rightarrow (x-1)(x+1) < 0$

$\Rightarrow x \in (-1, 1)$

$f'(x) > 0 \Rightarrow (x-1)(x+1) > 0$

$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

$f \begin{array}{c} \text{incr.} & \text{decr.} & \text{incr.} \\ -1 & & 1 \end{array}$

$f' \begin{array}{c} >0 & <0 & >0 \\ -1 & & 1 \end{array}$

f' changes from pos. to neg. at $x = -1$

So $f(-1) = 7$ is a local max

f' changes from neg. to pos. at $x = 1$. $f(1) = -1$ is a local min

(4)

Second Derivative Test

$$f''(x) = 20x^3$$

$$f'(x) = 0 \Leftrightarrow x = \pm 1$$

$$\text{So } f'(-1) = 0 \text{ and } f''(-1) = -20 < 0$$

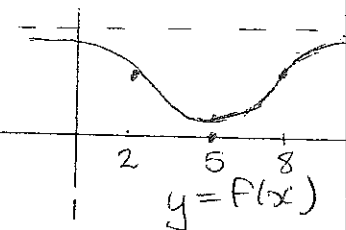
$\Rightarrow f$ is concave downward
Local max is $f(-1) = 7$

$$f'(1) = 0 \text{ and } f''(1) = 20 > 0$$

$\Rightarrow f$ is concave upward

Local min is $f(1) = -1$

29. $f'(5) = 0$
 $\Rightarrow$ horizontal
tangent at $x=5$



$f'(x) < 0$ when $x < 5$
 $\Rightarrow f$ is decreasing on $(-\infty, 5)$

$f'(x) > 0$ when $x > 5$
 $\Rightarrow f$ is increasing on $(5, \infty)$

$$f''(2) = 0 \quad f''(8) = 0$$

$$f''(x) < 0 \text{ when } x < 2 \text{ or } x > 8$$

$$f''(x) > 0 \text{ for } 2 < x < 8$$

$\Rightarrow f$ is concave upward on
 $(2, 8)$ and concave downward on $(-\infty, 2)$ and $(8, \infty)$

Inflection Points: $x=2$
 $x=8$

$$\lim_{x \rightarrow \infty} f(x) = 3 \quad y=3 \text{ is}$$

$$\lim_{x \rightarrow -\infty} f(x) = 3 \quad \text{HA}$$

even ~~40~~ $f(x) = x\sqrt{6-x}$

$$D_f: 6-x \geq 0 \Rightarrow x \leq 6$$

$$\text{So } D_f = (-\infty, 6]$$

$$\begin{aligned} a) \quad f'(x) &= \sqrt{6-x} + x \left(\frac{1}{2} (6-x)^{-\frac{1}{2}} (-1) \right) \\ &= \sqrt{6-x} - \frac{x}{2\sqrt{6-x}} \end{aligned}$$

$$= \frac{2(6-x) - x}{2\sqrt{6-x}} = \frac{12-3x}{2\sqrt{6-x}}$$

$$f'(x) > 0 \Rightarrow 12-3x > 0 \Rightarrow x < 4$$

$$f'(x) < 0 \Rightarrow x > 4 \text{ But } D_f = (-\infty, 6] \\ \therefore x \in (4, 6)$$

$$f' \quad \begin{array}{ccc} > 0 & & < 0 \\ & 4 & & 6 \end{array}$$

$$f \quad \begin{array}{ccc} \text{incr} & & \text{decr} \\ & 4 & & 6 \end{array}$$

(b) f' changes from pos to
neg at $x=4$

$$f(4) = 4\sqrt{2} \text{ is a local max}$$

No local min.

$$(c) \quad f'(x) = -\frac{3}{2} (x-4)(6-x)^{-\frac{1}{2}}$$

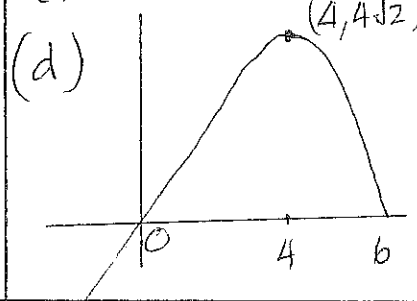
$$f''(x) = -\frac{3}{2} \left[(x-4) \left(-\frac{1}{2} (6-x)^{-\frac{3}{2}} (-1) \right) + (6-x)^{-\frac{1}{2}} (1) \right]$$

$$= \left(-\frac{3}{2} \right) \left(\frac{1}{2} \right) (6-x)^{-\frac{3}{2}} \left[(x-4) + 2(6-x) \right]$$

$$= \frac{3(x-8)}{4(6-x)^{3/2}}$$

$$f''(x) < 0 \text{ on } (-\infty, 6)$$

f is concave downward
on $(-\infty, 6)$ No Inflection
Point



B. Theory Questions (5)

1. (a) A critical number of a function is a number in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

b) A function is increasing on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .

2. (a) The Extreme Value Theorem

If f is continuous on $[a, b]$ then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at c and d in $[a, b]$.

(b) Fermat's Theorem

If f has a local maximum or minimum at c and if $f'(c)$ exists, then $f'(c) = 0$.

c) The Mean Value Theorem

Let f a function that satisfies the following hypotheses:

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b) .

Then there is a number $c \in (a, b)$ such that
$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or equivalently
$$f(b) - f(a) = f'(c)(b - a)$$

d) The Increasing / Decreasing Test. (I)

- If $f'(x) > 0$ on an interval then f is increasing on that interval I .

- If $f'(x) < 0$ on an interval I then f is decreasing on that interval I .

e) The Concavity Test

- If $f''(x) > 0$ for all x in I then the graph of f is concave upward on I .

- If $f''(x) < 0$ for all x in I then the graph of f is concave downward on I .

3. If $f' > 0$ on an interval ^(I) then f is increasing on that interval I .

Let x_1, x_2 be any two numbers on I with $x_1 < x_2$.

We need to show that $f(x_1) < f(x_2)$. Then f is increasing on I .

⑥

$f'(c) > 0$ given
So f is differentiable on interval I . Hence f is continuous on I (Theorem)

Since $x_1, x_2 \in I$ with $x_1 < x_2$
 f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2)

According to the Mean Value Theorem, there is a $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

- $f'(c) > 0$ (given and
- $x_2 - x_1 > 0$ because $x_1 < x_2$

Hence $f(x_2) - f(x_1) > 0$

So $f(x_1) < f(x_2)$

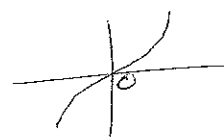
This shows that f is increasing on I .

2 (i) False

eg. $y = |x|$ in $x = 0$
 f is not differentiable in $x = 0$ *

(ii) False

eg. $y = x^3$ in $x = 0$
 $f'(x) = 0$ but f has no local extreme

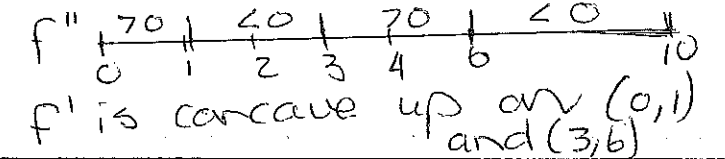
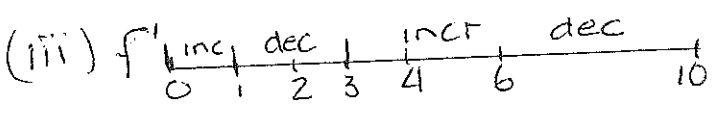


3. Consider graph of f'



(i) f is increasing on $(0, 2) \cup (4, 10)$

(ii) Local max is $f(2)$



4 $f(x) = \frac{\ln x}{x}$

(i) on $[1, 6]$ f is continuous
So Closed Interval method applies.

(ii) • Critical numbers

$$f'(x) = \frac{(\frac{1}{x})x - (\ln x)(1)}{x^2} = \frac{1 - \ln x}{x^2}$$

$$f'(x) = 0 \text{ if } 1 - \ln x = 0 \Rightarrow \ln x = 1 \Rightarrow x = e \in D_f$$

$f'(x)$ DNE if $x = 0 \notin D_f$

Only critical number $c = e$

x	1	e	6
$f(x)$	0	0.37	0.3

Absolute max $f(e) = 0.37$
Absolute min $f(1) = 0$

C. Test Questions

1 (i) $f(x) = \sqrt{x}$ $D_f = [0, \infty)$
 $f'(x) = \frac{1}{2\sqrt{x}}$

f' does not exist if $x = 0$

Critical number $c = 0$

(ii) $f(x) = \ln x$ $D_f = (0, \infty)$
 $f'(x) = \frac{1}{x}$ does not exist if $x = 0$.

BUT $x \notin D_f$

So $x = 0$ is not a critical number